Cyber Security Minor Project: Kali Linux OS

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1. Introduction

Kali Linux is a operating system that is build for cyber security and penetration testing. It have many tools which is very usefull for hackers and security expert. It is essential in cybersecurity domain because it give us safe place to test vulnerability without risk.

2. Objective

The main goals for this minor project is to successfully installing Kali Linux in a virtual environment. Also, we will performing system updates to keep the machine secure from known bugs. After that, we doing MAC address spoofing to hide our device identity on the network, and finally manage user accounts for better security and access control.

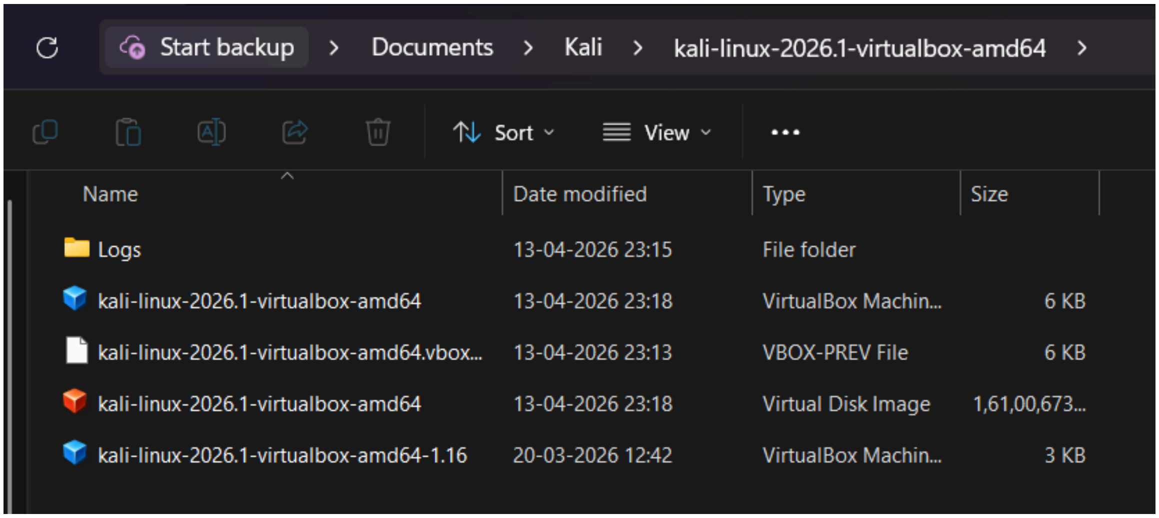
3. Setup

For the hardware prerequisites, the host machine need a minimum of 8GB RAM, at least 20GB of free hard disk space, and a multi-core processor with virtualization enabled in the BIOS. For software, we are using Oracle VirtualBox. We need virtualization because it provide a safe, isolated container. If we make mistake and crash the Kali system during testing, our main host operating system will not getting damaged. Virtualization is needed because we not want to damage our main host computer, so tools like VirtualBox help to make isolated virtual environment.

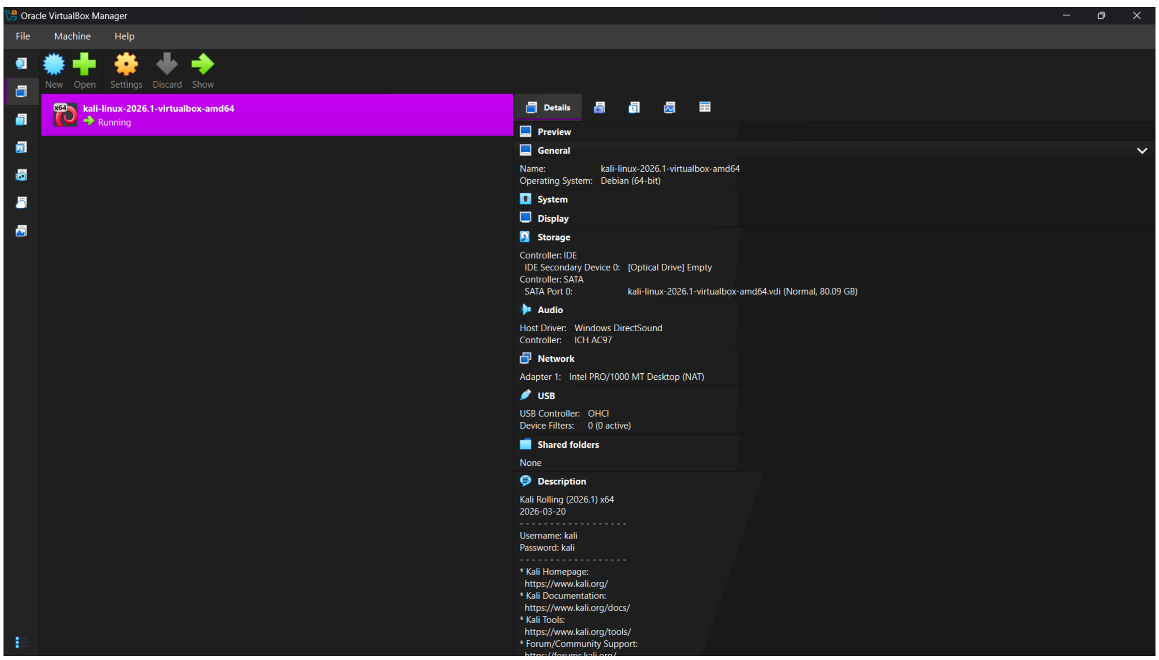
4. Tasks

Installation

I download pre-build VirtualBox image of Kali linux from official site. Then I import it in VirtualBox, set network to NAT, and start it. The default login user is kali/kali.



Start backup > Documents > Kali > kali-linux-2026.1-virtualbox-amd64 >
Sort
View
Name
Date modified
Type
Size
Logs
13-04-2026 23:15
File folder
kali-linux-2026.1-virtualbox-amd64
13-04-2026 23:18
VirtualBox Machin...
6 KB
kali-linux-2026.1-virtualbox-amd64.vbox...
13-04-2026 23:13
VBOX-PREV File
6 KB
kali-linux-2026.1-virtualbox-amd64
13-04-2026 23:18
Virtual Disk Image
1,61,00,673...
kali-linux-2026.1-virtualbox-amd64-1.16
20-03-2026 12:42
VirtualBox Machin...
3 KB



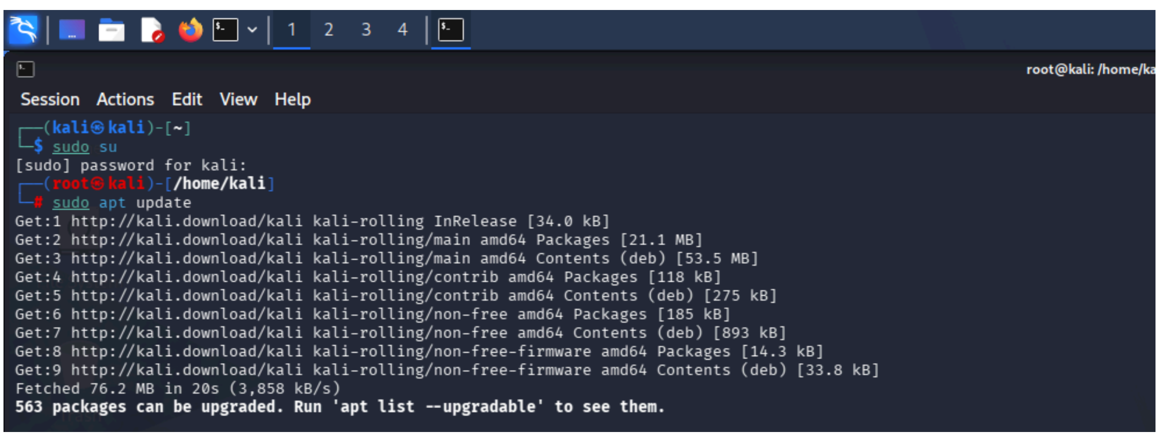
Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Show
kali-linux-2026.1-virtualbox-amd64
Running
Details
Preview
General
Name: kali-linux-2026.1-virtualbox-amd64
Operating System: Debian (64-bit)
System
Display
Storage
Controller: IDE
IDE Secondary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: kali-linux-2026.1-virtualbox-amd64.vdi (Normal, 80.09 GB)
Audio
Host Driver: Windows DirectSound
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB
USB Controller: OHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
Kali Rolling (2026.1) x64
2026-03-20
- - - - - - - - - - - - - - - - -
Username: kali
Password: kali
- - - - - - - - - - - - - - - - -
* Kali Homepage:
https://www.kali.org/
* Kali Documentation:
https://www.kali.org/docs/
* Kali Tools:
https://www.kali.org/tools/
* Forum/Community Support:

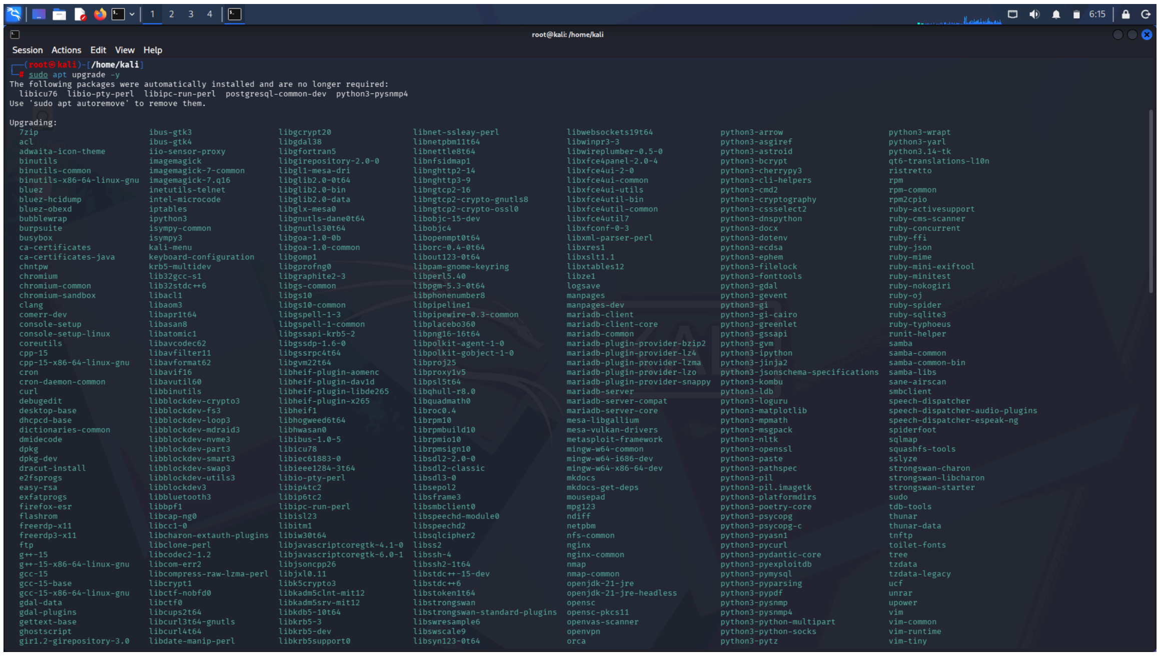
Update and Upgrade

To update system I open terminal and type command:

sudo apt update && sudo apt upgrade -y

This download and install latest packages.



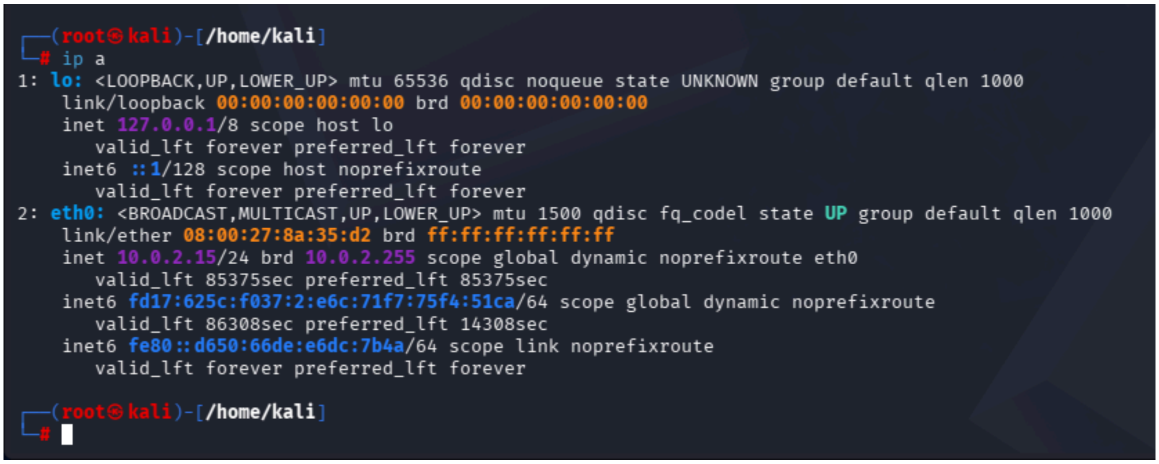


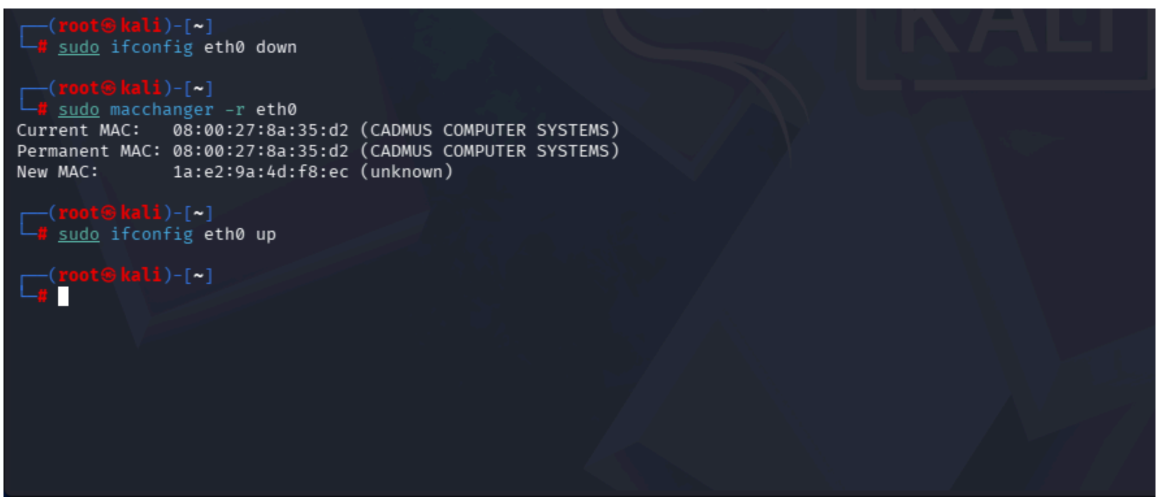
MAC Address Spoofing

To changing our MAC address, we need a tool called `macchanger`. Changing the MAC address hide our physical hardware address from the network router. The steps I followed in terminal was:

1. *Find interface: ip a*
2. *Bring interface down: sudo ifconfig eth0 down*

3. Change the MAC to a random one: sudo macchanger -r eth0
4. Bring interface up: sudo ifconfig eth0 up

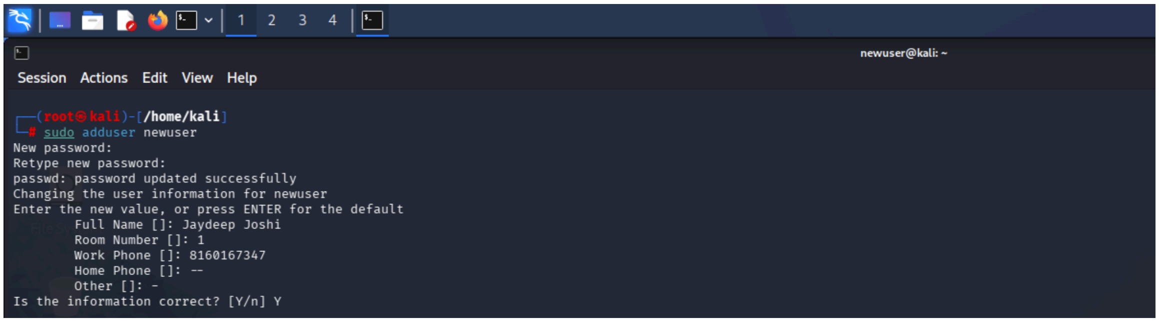


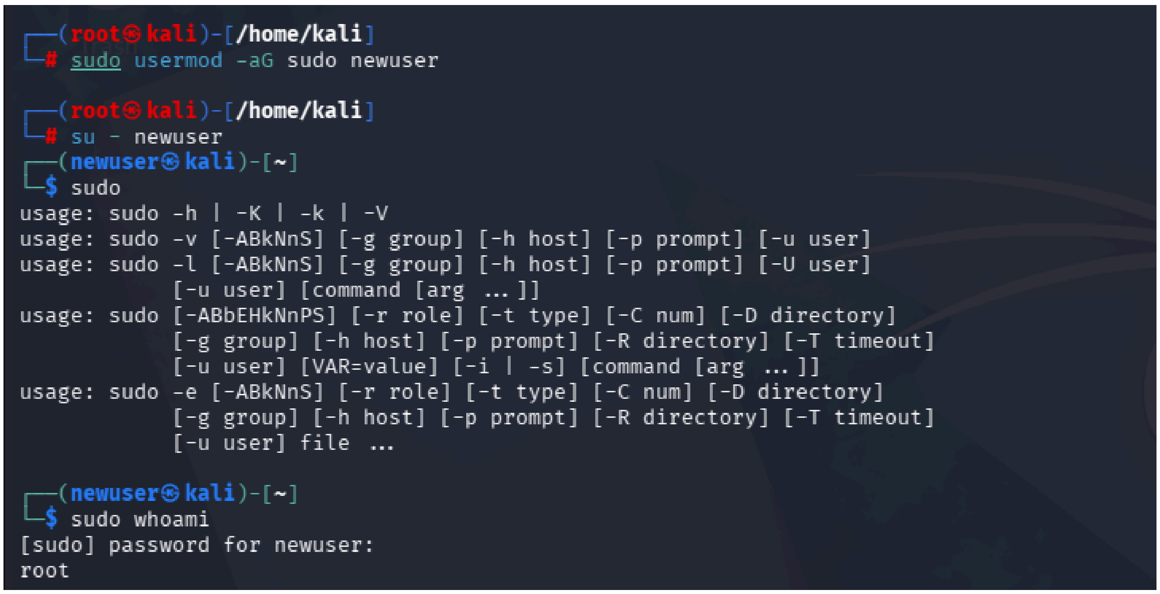


User Management

Working as the root user all the time is very dangerous because one wrong command can break the system. So, I added a new user for daily tasks. I used this commands in terminal:

1. Add user: sudo adduser internuser
2. Assign admin: sudo usermod -aG sudo internuser
3. Switch user: su - internuser
4. Sudo whoami:- to verify admin access





5. Documentation

Each task is essential in cybersecurity operations. Doing update keep system safe from new bugs. MAC spoofing hide our real physical device address on network which is important for privacy during testing. Good user management stop unauthorized peoples from getting root access.

6. Reflection

In conclusion, this project teach me how to setting up a professional cyber lab from scratch. I learned that virtualization is a safe, flexible way to practice hacking skills. This skills applies directly to real-world scenarios, like when we have to test a client's network. We must spoof our MAC so we don't trigger defensive alarms, and we must secure our own machines with proper user accounts before we start attacking others. It is very good preparation for my upcoming internship and my future career.